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Predicting Concurrent and Short-Term Desire and Intent to Attempt Suicide Among People with Body Dysmorphic Disorder Using Ecological Momentary Assessment of Anxiety and Shame

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Abstract

Anxiety and shame are central, elevated emotions in body dysmorphic disorder (BDD) that are implicated as risk factors for suicide in suicide theories, and which are associated with suicide risk in cross-sectional BDD studies. Given that emotions are transient and suicide risk can increase quickly, risk prediction in BDD may be enhanced by measuring momentary anxiety and shame. In 87 adults with moderate to severe, clinician-diagnosed BDD, we collected ecological momentary assessment (EMA)-rated anxiety and shame three times daily for two 14-day periods (28 days). We used generalized linear mixed models to estimate associations of concurrent or next-observation EMA-rated (1) intensity of desire to die by suicide and (2) intention (absent/present) to attempt suicide, above baseline clinician-assessed suicide ideation (SI) severity (Columbia-Suicide Severity Rating Scale). Higher within-person deviations from one's average anxiety and shame were significantly related to greater concurrent and next-observation suicide desire and concurrent intent, above baseline clinician-assessed SI severity. Only baseline clinician-assessed SI predicted next-observation suicide intent. Altogether, results showed that our ability to detect and predict suicide risk in BDD was improved beyond baseline clinician assessment when a person's current anxiety and shame levels were also considered. When someone with BDD experiences elevated anxiety or shame compared to their own norm, these elevations are associated

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Transparency and Openness: This study was approved by the Mass General Brigham IRB (protocol #2019P002041, study title: Understanding daily changes in BDD risk using smartphones). This study is a secondary analysis; the primary study's design was pre-registered at [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT04254575) (NCT04254575). Analysis code and output are available as supplementary materials. Analyses were conducted using R, version 4.4.1 (R Core Team, 2020). Data are not available because participants did not consent to data sharing. Results related to inter-rater reliability of study diagnostic assessments were previously presented in an earlier study manuscript (Weingarden et al., 2025); no other results have been presented previously. No part of this article was written by artificial intelligence. We report power analysis, all data exclusions (if any), manipulations, and measures used in the study.

Trial Registration: [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT04254575) Identifier: NCT04254575

with concurrent and short-term increases in suicide desire and concurrent suicide intent. This is the first prospective study of shame and anxiety as risk factors for SI in BDD. Results underscore the importance of these emotions as assessment and intervention targets.

General Scientific Summary:

This study suggests that when people with body dysmorphic disorder experience higher than their usual levels of shame or anxiety, they may be more likely to experience suicide desire (at that moment or soon thereafter) or suicide intent (at that moment). Results highlight that heightened shame and anxiety can be a useful real-time clue – for both patients and their clinicians – that one’s suicidal thinking may worsen, giving the patient and their clinician an opportunity to use strategies to manage anxiety and shame with the aim of preventing worsening suicidal thoughts.

Keywords

Body dysmorphic disorder (BDD); ecological momentary assessment (EMA); anxiety; shame; suicide risk

Body dysmorphic disorder (BDD) is a psychiatric illness that involves a persistent, distressing belief that one’s physical appearance is flawed or defective, accompanied by repetitive, time-consuming rituals intended to conceal or improve perceived appearance flaws (American Psychiatric Association, 2013). BDD occurs in 1.7-2.9% of the general population (Buhlmann et al., 2010; Koran et al., 2008; Rief et al., 2006; Schieber et al., 2015).

Suicide risk in BDD is notably elevated. Most (78%) people with BDD experience suicidal thoughts (Phillips & Menard, 2006) and between one-quarter to half make a suicide attempt (Addison et al., 2024; Phillips et al., 2005, 2007; Phillips & Menard, 2006; Veale et al., 1996). Recent nationwide Swedish data show that rates of completed suicide in BDD are 3 times higher than in an age-, sex-, and geographically-matched general population cohort (Rautio et al., 2024). Despite pronounced suicide risk, relatively limited research has explored reasons for high rates of suicidal thoughts and behaviors in BDD.

Several eminent theories of suicide converge in the idea that suicide is a means to escape especially painful emotions. Baumeister’s (1990) Escape Theory of suicide posits that an individual can become suicidal when they perceive that they fall short of personal or societal standards (an experience that is common to those with BDD), attribute that shortcoming to a personal failure, experience painful emotions such as shame in response, and seek to escape the emotional pain via suicide. The Integrated Motivational-Volitional (IMV) model of suicide emphasizes the experience of entrapment – that is, feeling defeated, humiliated, and lacking escape – underscored by shame and anxiety – as motivating individuals to consider suicide (O’Conner & Kirtley, 2018). Models of acute suicide risk (i.e., Acute Suicidal Affective Disturbance [Tucker et al., 2016; Rogers et al., 2017], Suicide Crisis Syndrome [Galynker et al., 2017]) note that sudden increases in anxiety and self-hatred or shame and their behavioral counterparts of avoidance and social withdrawal put people at acute risk for suicide (Rogers et al., 2017). Empirical evidence also supports theoretical

frameworks positing that suicide is often motivated by a desire to escape painful emotions, especially including anxiety and shame (Nock et al., 2025; Bryan et al., 2013).

Anxiety and shame are prominent and elevated in BDD, highlighting these emotion states as candidate suicide risk factors in this population. Anxiety is a high arousal negative emotion that involves tension and worry when perceiving a distal threat (Kazdin, 2000), leading to behavioral avoidance and maladaptive safety behaviors such as rituals. Shame is a deeply painful self-conscious emotion that is experienced when a person judges themselves in a global and harshly critical manner (e.g., “I am defective,” “I am bad”), accompanied by social withdrawal and isolation (Tangney & Dearing, 2002). In people with BDD, anxiety arises frequently in the context of maladaptive self-focused attention and social anxiety tied to one’s appearance concerns (“everyone is staring at me and judging me negatively because of my ugly nose”) (Fang & Hofmann, 2010). Shame is experienced almost invariably in BDD, when people overvalue the importance of physical appearance and overgeneralize the meaning of having a (perceived) appearance defect (“if I’m ugly, then I’m unlovable / defective”) (Weingarden & Renshaw, 2015). Cognitive-behavioral conceptual models of BDD underscore anxiety, shame, and their behavioral responses (i.e., avoidance, social withdrawal) as central to BDD maintenance and risk for suicide (Angelakis et al., 2016; Fang & Wilhelm, 2015; Feusner et al., 2010; Weingarden & Renshaw, 2015). Consistent with theory, empirical research also shows that anxiety and shame are significantly elevated among individuals with BDD compared to nonpsychiatric controls and psychiatric comparisons (Weingarden et al., 2016, 2018).

Initial empirical research also supports this relationship between anxiety, shame, and risk for suicide in BDD. In cross-sectional BDD studies, moderate to strong relationships exist between anxiety and suicide risk ($r = .52$) (Weingarden et al., 2016), shame and suicidal ideation ($r_s = .39$ to $.48$) (Weingarden et al., 2016, 2018), and shame and hopelessness ($r = .50$) (Weingarden et al., 2018). Additionally, comorbid anxiety disorders are associated with significantly increased likelihood of self-harm and suicide attempts (OR: 2.05, $r = .19^a$) in people with BDD (Addison et al., 2024). Moreover, reductions in shame across serotonin reuptake inhibitor treatment for BDD were significantly associated with reductions in suicidal thoughts (Beta = .39) and hopelessness (Beta = .41), even when accounting for reductions in BDD severity and depression severity (Weingarden et al., 2018). Taken together, evidence suggests that anxiety and shame are associated with elevated suicide risk in BDD.

However, prior research has used either single time point cross-sectional or pre-post treatment designs to assess relationships between anxiety and shame with suicidal thoughts and behaviors in BDD – designs which preclude understanding the short-term temporal dynamics between variables. Because negative emotions and suicidal thoughts vary widely in short time frames, and risk for suicide often develops quickly (Kleiman et al., 2017), it is critical to use methods that capture momentary associations between shame and anxiety with

^aTo facilitate comparison across effect sizes noted in our introduction, we converted this effect size using an online effect size conversion calculator (Lin, n.d.)

suicide risk, and which test whether anxiety and shame are predictive of suicide risk over short periods of time.

A small number of studies outside the domain of BDD have used ecological momentary assessment (EMA) to examine concurrent or prospective relationships between anxiety or shame and suicide risk. Mou et al. (2018) showed that 10 EMA-rated negative affect states, including “anxious,” ($r = 0.84^a$) were each associated with concurrent EMA-rated suicidal ideation in 35 adult psychiatric inpatients. Similarly, Forkmann and colleagues (2018) showed that EMA-rated anxiety was significantly associated with concurrent EMA-rated passive ($r = .46$) and active ($r = .39$) suicidal ideation (SI) in a psychiatric inpatient sample with depressive disorders ($N = 74$). Husky and colleagues (2017) showed that EMA ratings of anxiety predicted subsequent SI in 42 adults post-discharge from hospitalization for suicide risk. Finally, in 104 adult inpatients who had made recent suicide attempts, Bentley and colleagues (2021) showed that mean levels of shame/self-hatred and anxiety/agitation during hospitalization predicted the number of days patients experienced SI after discharge; additionally, mean and variability of shame/self-hatred (but not anxiety/agitation) predicted post-discharge suicide attempts (Bentley et al., 2021).

Altogether, there is a clear need for research within BDD that extends single time-point studies, to examine momentary cross-sectional and short-term prospective relationships between anxiety and shame with suicide risk. Results that link anxiety and shame temporally with concurrent and subsequent suicide risk – especially if negative emotions predict risk above and beyond gold-standard clinician assessment of risk at baseline – would underscore the value of timely assessment and interventions that target anxiety and shame as they arise, to reduce acute risk. To this end, the aim of the present study was to examine whether EMA-rated anxiety and shame predict same- and next-observation EMA-rated intensity of suicidal desire and presence of suicidal intent in a BDD sample, above and beyond clinician-assessed past month severity of SI collected at baseline via the Columbia-Suicide Severity Rating Scale (C-SSRS).

Method

Participants

Eligible participants had a current primary psychiatric diagnosis of BDD based on the Mini International Neuropsychiatric Interview (MINI 7.02), with moderate to severe BDD symptoms based on a Yale-Brown Obsessive Compulsive Scale Modified for BDD (BDD-YBOCS) score ≥ 24 , in addition to (a) thoughts of suicide in the past month, (b) alcohol use in the past 2 weeks, and/or (c) marijuana use in the past 2 weeks. Eligibility criteria related to alcohol and marijuana use were part of broader primary aims of the study and are not directly related to the present secondary analysis. Inclusion criteria required that participants own a smartphone (Android or iOS) and computer or tablet and have Wi-Fi access at home, to complete study procedures. Participants were excluded if they had current or recent (past 6 months) mania or hypomania assessed via the MINI; suicide risk or BDD severity warranting higher than outpatient care, defined by clinician judgment and a score ≥ 4 on item 1 of the C-SSRS, which assesses suicidal ideation; or insufficient technology literacy to complete study procedures.

We recruited participants via referrals (i.e., within our clinic and research program), via advertisements and postings on BDD organizations' (e.g., International OCD Foundation) websites and newsletters, and via advertisements on social media sites and online community groups, largely including Reddit communities focused on appearance concerns or mental health.

Procedures

At baseline, participants completed clinician-administered diagnostic and clinical assessments, as well as self-report demographic, clinical, and psychosocial surveys. In two separate 2-week periods occurring after the baseline appointment and after the midpoint assessment 1.5 months later, participants completed EMA questionnaires (see EMA, below, for details). Additional self-report and clinician-administered data were collected at midpoint (1.5 months after baseline) and endpoint (3 months after baseline) assessments, and passive smartphone sensor data were collected for approximately 3 months; these data are not used in the present study. Data were collected between July 2020 and May 2023.

Pre-Screening: Individuals interested in study participation were invited to schedule a phone call with the study coordinator, who gave information about the study and pre-screened participants for likely eligibility.

Virtual Screening and Baseline Visit: Individuals deemed potentially eligible after the phone call with the study coordinator completed a virtual (i.e., via HIPAA-compliant video conference) screening and baseline visit with a study clinician, who was either a licensed psychologist specialized in BDD or a MA-level pre-doctoral fellow in the same BDD specialty center. During the screening visit, the study clinician obtained informed consent and subsequently administered clinician-rated assessments to evaluate psychiatric diagnoses, BDD severity, and suicidal thoughts and behaviors that they were trained to deliver (see Measures). Assessments were audiotaped such that 15% could be randomly selected for rating by an independent trained and reliable assessor to obtain inter-rater reliability estimates for primary diagnostic (MINI) and severity (BDD-YBOCS) measures. Inter-rater reliability was strong (intraclass correlation [2,1] for BDD-YBOCS = .93, 100% agreement on MINI BDD diagnosis) (Weingarden et al., 2025). Eligible individuals then met virtually with the study coordinator to download and set up the study MetricWire app (Metricwire, 2023) for EMA data collection. Following the virtual visit, participants completed baseline self-report surveys online via REDCap (Harris et al., 2009, 2019).

Ecological Momentary Assessments (EMA): Participants completed EMA questionnaires in two separate 2-week periods following the baseline and midpoint timepoints, respectively (28 days in total). These two periods were separated by 1.5 months, to minimize participant burden and maximize adherence and variability of experiences reported. See EMA Measures for additional details.

For each day during an EMA period, participants were prompted to complete four questionnaires in total. One served as a daily diary and was available for completion at any point within a 7am-11am time frame, which asked participants to report how they were

feeling and what they were experiencing “overall yesterday.” Data from this questionnaire were not included in the present study. The other three questionnaires were sent at random times within fixed time frames: (1) 10:00am-1:30pm, (2) 1:45pm-5:15pm, and (3) 5:30pm-9:00pm. Participants had 30 minutes from the time of prompting to complete a questionnaire, before it expired. Each of these questionnaires asked participants to report how they were feeling and what they were experiencing “right now.”

Measures

Clinician-Administered Measures: The MINI (Sheehan et al., 1998) is a well-validated semi-structured diagnostic interview; it correlates strongly with the Structured Clinical Interview for DSM (SCID) and takes less time to administer (Sheehan et al., 1997). The BDD-YBOCS (Phillips et al., 1997) is a semi-structured assessment of BDD severity in the past week that ranges from 0-48 (higher scores indicate higher severity). It displays excellent inter-rater and test-retest reliability, strong internal consistency, and good convergent and discriminant validity with other mental health assessments (Phillips et al., 1997, 2001). The Columbia-Suicide Severity Rating Scale (C-SSRS) (Posner, 2016) is a semi-structured evaluation of suicidal ideation and behaviors over one’s lifetime and past month; it displays high sensitivity and specificity for suicidal behavior, moderate to strong internal consistency, and good convergent and discriminant validity with other assessments of suicidality (Posner et al., 2011). Scores from item 1 of the C-SSRS, which capture past month severity of suicidal ideation, were included as a predictor in all statistical models, to account for gold-standard, clinician-assessed severity of suicidal ideation.

Self-Report Questionnaires: Participants completed a self-report demographics survey at baseline.

EMA Measures: Items from the Positive and Negative Affect Scale - X (PANAS-X) (Haney et al., 2023; Watson et al., 1988) were used to assess current intensity of a range of emotions (“indicate to what extent you feel this way right now”) on a scale from 1 (“very slightly or not at all”) to 5 (“extremely”). The PANAS-X is a well-validated scale that reliably detects between-person differences and within-person change in the context of EMA (Haney et al., 2023). We added an “anxious” item to assess current anxiety, and the “ashamed” item was used to assess current shame.

We assessed current intensity of suicide desire and intent using items developed by Kleiman and colleagues (Kleiman et al., 2017), asking participants “How intense is your desire to kill yourself right now?” and “How strong is your intention to kill yourself right now?”. The scales for both items range from 0 (“not intense/strong at all”) to 10 (“very strong”).

Data Analysis

Overview and Computing Key Variables: We separated EMA-rated anxiety and shame into between-person and within-person components, such that between-person anxiety and shame were participant means for each variable across all observations for that individual relative to the sample mean, and within-person anxiety and shame were each observation’s deviation from that participant’s mean (i.e., momentary variation from one’s

average emotion level). We calculated the time (hours) between each successive observation, to include as a covariate in prospective models. To aid results interpretation, we rescaled independent variables so that means were 0 and standard deviations were 1. As 95.86% of responses to EMA-rated intent to kill oneself were 0s, 1s or 2s, we computed a binary intent variable (where a score of 0 represented absent intent, and a score >0 represented some intent) to use in models predicting same and next-observation suicide intent.

Modeling Approach: All models were generalized linear mixed models (GLMMs), modeling three times daily within-person (i.e., person-mean centered) anxiety and shame and between-person (i.e., grand mean centered) anxiety and shame as predictors of dependent variables (i.e., intensity of suicide desire, suicide intent [absent/present]), adjusting for grand mean-centered baseline suicide ideation severity (indexed via C-SSRS). We included a by-subject random intercept and slope across EMA day to account for correlated ratings during the observational period. As a covariate in prospective models, we included the time (hours) between each successive observation to account for variability in when participants responded to EMA prompts. GLMMs used a gamma likelihood for skewed distributions (suicidal desire) and a binomial likelihood for dichotomous data (suicidal intent). Power analysis for a two-level multilevel model using the EMAtools R package (Kleiman, 2017) accounting for 87 participants, 28 days of surveys, and 3 surveys per day, an underestimated 50% survey completion rate across participants, and an ICC estimate of 0.5 suggests power of roughly 1.0 to detect medium effects ($d=.5$) and 0.9 to detect small effects ($d=.2$).

Adjusting for False Discovery Rate: We adjusted p values using the Benjamini-Yekutieli (BY) procedure (Benjamini & Yekutieli, 2001) to control for false discovery rate, given multiple hypothesis testing with dependent data.

Concurrent Models: In two separate models, we tested whether (a) intensity of desire to die by suicide or (b) intention to die by suicide was associated with one's concurrent state (within-person) and average (between-person) levels of anxiety and shame above baseline clinician-assessed suicide ideation severity.

Prospective Models: In two separate models, we tested whether *future* (a) desire to die by suicide and (b) suicide intent was associated with one's prior (last observation) state and average levels of anxiety and shame above baseline clinician-assessed suicide ideation severity.

Transparency and Openness

This study is a secondary analysis; the primary study's design was pre-registered at [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT04254575) (NCT04254575). Results related to inter-rater reliability of study diagnostic assessments were previously presented in an earlier study manuscript (Weingarden et al., 2025); no other results have been presented previously. Analysis code and output is available as supplementary materials. Analyses were conducted using R, version 4.4.1 (R Core Team, 2020). The data that support the findings of this study are not

publicly available due to their sensitive nature and because study participants did not consent to data sharing.

Results

Sample Characteristics

The present sample includes all consented and eligible participants who completed *any* EMA questionnaires ($N = 87$), as analytic models are robust to missing data (Little, 1995). This approach follows similar procedures to prior EMA studies (e.g., Armey et al., 2018).

Participants had a M (SD) age of 29.39 (8.99) years (range: 18–55 years) and severe BDD symptoms on average (BDD-YBOCS $M = 30.78$, $SD = 3.62$). Almost all (85/87, 97.70%) participants endorsed lifetime passive or active SI on the baseline C-SSRS, and 65/87 (74.71%) endorsed some passive or active SI in the past month on the baseline C-SSRS. Figures 1 and 2 show distributions of participants' non-zero EMA responses to items about current intensity of desire to kill oneself and intent to kill oneself, respectively. Most participants experienced some suicide desire and intent in EMA items across the study period, with only 14.94% never endorsing suicide desire scores >0 , and 40.23% never endorsing suicide intent scores >0 . Mean (SD) daily EMA-reported suicide desire and intent scores (per participant per day) were 1.37 (1.94) and 0.42 (0.87), respectively. Ratings of 0 on EMA measures of suicide desire and intent comprised 56% and 77% of total EMA responses to these items, respectively. Table 1 describes other key demographic and clinical characteristics of the sample.

Overall, EMA compliance was strong; out of 84 possible prompts, participants completed a mean of 61.25 (72%), $SD = 16.21$ (19%), and 80.46% of participants completed $>60\%$ of all EMA prompts. We used Spearman's rho to assess collinearity between EMA-rated anxiety and shame. Results ($\rho = .51$, $p < .001$) suggested moderate correlation, but not multicollinearity (e.g., $\rho \geq 0.9$) (Siegel, 1988) between measures, supporting inclusion of both emotions as separate independent variables in models.

Concurrent Models

See Table 2 for results from concurrent models of desire and intent to die by suicide.

Suicide Desire: Higher past-month suicidal ideation severity at baseline (C-SSRS) and higher within-person deviations from one's own average levels of anxiety and shame were significantly associated with greater concurrent suicidal desire. A 1 SD increase in within-person anxiety and shame was associated with a 12% and 14% increase in concurrent suicide desire, respectively.

Suicide Intent: Higher past-month suicidal ideation severity at baseline (C-SSRS) and higher within-person deviations from one's own average levels of anxiety and shame were significantly associated with suicide intent. A 1 SD increase in within-person anxiety and shame was associated with 1.58 and 1.61 times the odds of reporting suicide intent, respectively.

Prospective Models

See Table 3 for results from prospective models of desire and intent to die by suicide.

Suicide Desire: Higher past-month suicidal ideation severity at baseline (C-SSRS) and higher within-person deviations from one's average levels of anxiety and shame significantly predicted greater next-observation intensity of suicide desire. A 1 *SD* increase in within-person anxiety and shame were both associated with a 4% increase in concurrent suicide desire.

Suicide Intent: Only higher past-month suicide ideation severity at baseline (C-SSRS) was significantly associated with next-observation suicide intent.

Discussion

Risk for suicide in BDD is notably high (Rautio et al., 2024) and can rapidly fluctuate (Kleiman et al., 2017). Clinicians who treat BDD are tasked with assessing risk as effectively as possible. At present, however, our ability to accurately predict suicide across populations is relatively poor (Kessler et al., 2019). Frontline clinical practice typically involves using validated clinical interviews such as the C-SSRS, which are administered at maximum at standard clinical visits (e.g., weekly), due to their administration burden. Given that suicide risk often develops over short periods of time such as hours or days, risk assessment may be meaningfully improved with more frequent, and more focused, assessment of well-conceptualized risk factors. Eminent theories of suicide (e.g., Baumeister, 1990; O'Connor & Kirtley, 2018) posit that suicidal thoughts and behaviors arise as a means of escaping from painful emotions - particularly anxiety and shame. In BDD, both theory and cross-sectional research also point to anxiety and shame as promising variables that - when assessed simply and frequently - may enhance short-term prediction of suicide risk even above less frequently-collected, gold-standard clinician-assessed risk.

To this end, the present study extends past single timepoint and pre/post BDD study designs (Addison et al., 2024; Weingarden et al., 2016, 2018). It is the first to use EMA to study hypothesized suicide risk factors over short-term time frames in BDD, and it is the first study to examine prospective relationships between shame and anxiety with suicidal thoughts in BDD. Methodological strengths include a relatively large ($N=87$), well-characterized, clinician-diagnosed BDD sample with high SI prevalence and strong EMA compliance over 84 prompts and 28 days. Results showed that within-person anxiety and shame were associated with current and next-observation severity of suicide desire, and current presence of suicide intent, above baseline clinician-assessed SI severity. In other words, when an individual's anxiety or shame was higher than their own average, their concurrent intensity of suicide desire and suicidal intent was significantly higher, as was their future (next-observation) suicide desire, but not intent. Effect sizes highlight that within-person deviations in anxiety and shame are similarly strong risk factors.

Interestingly, while elevations from a person's norm were associated with acute suicide risk, an individual's *average* (i.e., between subjects) level of anxiety or shame was *not* significantly associated with risk above and beyond other independent variables. These

findings once again point to the value of momentary assessment that reveals how a person is feeling right now *relative* to their norm, as opposed to one's average level, which is typically gathered via imprecise retrospective assessment methods.

Results underscore ways to enhance risk assessment and intervention with BDD patients. First, across all models, baseline clinician-assessed SI severity on the C-SSRS was significantly associated with suicidal thoughts, underscoring the importance of clinician risk assessment as a starting point. Building on this foundation, findings suggest that both anxiety and shame should be considered during risk assessment and intervention in BDD; they are theoretically distinct, statistically non-collinear, contribute additive value in models of suicide desire and concurrent intent, and respond to somewhat different intervention strategies. To this latter point, when a patient with BDD is experiencing risk for suicide associated with a spike in anxiety, the clinician may use cognitive restructuring (e.g., to reduce catastrophizing, black-and-white thinking, mindreading) in the short term; to reduce the long-term occurrence of anxiety-based suicide risk, exposure with response prevention may be a frontline choice. When helping a patient manage risk for suicide associated with acute shame, they may emphasize compassion-focused skills and, in the long-term, target deeper-level core beliefs. When assessing for risk, clinicians (alongside their patients) should assess current levels of anxiety and shame and also consider whether patients *anticipate* facing stressors likely to induce higher-than-usual anxiety or shame, as this would indicate reason to be on alert for changes in suicide risk. For example, an upcoming work deadline or potential interpersonal rejection might lead to a temporary spike in anxiety or shame and, correspondingly, suicide desire or intent as a way to escape these difficult emotions (Baumeister, 1990; Bryan et al., 2013; Nock et al., 2025). Assessing for upcoming situations likely to trigger anxiety or shame would inform clinicians that continued risk assessment is warranted during the time frame of the stressor. Importantly, educating patients to think ahead about likely triggers of anxiety and shame can also help *patients* learn to anticipate and prepare for potential changes in risk (e.g., with a plan to use CBT skills that target anxiety and shame, with up-to-date safety plans), and it can help patients build self-awareness about these patterns (e.g., observe that anxiety and shame – as well as suicide desire and intent – can ebb and flow quickly, giving an opportunity to ride out peaks with the knowledge that these difficult emotions will pass).

In the absence of a clinical comparison group, we cannot know whether results and their corresponding clinical implications are specific to BDD, or whether they would generalize to other populations at risk for suicide who experience high levels of anxiety and shame. Indeed, our interest in studying anxiety and shame as suicide risk factors stemmed *both* from broader suicide theories *and* from cognitive-behavioral models of BDD specifically, which note that these two emotions are pronounced, characteristic experiences in BDD – a disorder with notably elevated risk of suicide. To this end, we expect that findings are especially relevant to those with BDD, and we anticipate that results would extend to other populations for which anxiety and shame are also core, prominent experiences. In support of this latter point, in a cross-sectional study of nonclinical adults ($n = 133$), adults with self-reported BDD ($n = 114$), and adults with self-reported obsessive compulsive disorder (OCD) ($n = 114$), shame was significantly, similarly associated with suicidal ideation among those with OCD and BDD and not associated with suicidal ideation in healthy controls; a similar but

non-significant pattern emerged for anxiety (Weingarden et al., 2016). The broader suicide literature has also found links between anxiety and shame with suicide risk in other forms of psychopathology [e.g., Bentley et al., 2021; Forkmann et al., 2018; Huskey et al., 2017; Mou et al., 2018]), corroborating the potential generalizability of these findings outside of BDD.

Findings should be interpreted considering some study limitations. First, our study focused on predicting thoughts of suicide, which are distinct from and much more common than suicide behaviors such as attempts. The present study therefore does not provide data on the role of shame or anxiety as risk factors for suicide behaviors, which are ultimately the most clinically salient outcomes to predict. Second, while suicidal thoughts were prevalent and varied in the sample (e.g., 85.06% endorsed suicide desire on EMA questionnaires, 59.77% reported suicide intent on EMA questionnaires) and risk-related exclusion criteria were less stringent than in published digital intervention studies for BDD (Jaroszewski et al., 2025), we excluded and provided referrals to five participants who, at screening, were experiencing active suicidal ideation with a plan or intent to act in the past month (≥ 4 on the C-SSRS suicidal ideation severity item). We carefully designed eligibility criteria related to suicide risk, with the goal of balancing external validity, participant safety, and feasibility / person power. We chose to exclude participants on the basis of current active SI with intent or plan given the complexity of managing imminent suicide risk in a nationally recruited sample, and to balance feasibility of implementing the safety protocol, which included telephone risk assessments in response to very high scores on EMA suicide items (i.e., 10 for desire, 8+ for intent). However, given this exclusion criterion, results may not generalize to those who are experiencing active SI with intent or plans. Relatedly, to accommodate a zero-inflated distribution of scores on our EMA item assessing intensity of suicide intent, we converted it into a binary variable that coded absence versus presence of any suicidal intent, thereby sacrificing some variability in this important dependent variable. That said, scores > 0 comprised 23% of scores on the intent variable, and 95.86% of scores fell in the range of 0, 1, or 2; the relatively restricted range of scores > 0 likely reduced the impact of the choice to dichotomize the intent variable. Future studies that include participants experiencing the full range of suicide desire and intent – and which extend this study to also predict suicide behaviors – will build on present findings. Eligibility criteria related to alcohol and marijuana use were included due to broader primary aims of the study but were not directly related to the present secondary analyses. Given that alcohol and substance use are common behaviors in those with BDD (e.g., substance use disorders in 49% [Grant, Menard, Pagano, Fay, & Phillips, 2005]), we expect a relatively attenuated impact of these eligibility criteria on sample generalizability.

Notwithstanding these limitations, the present study is the first to build on exclusively cross-sectional work, to demonstrate that anxiety and shame are *prospective risk factors* for suicide desire in BDD. The study offers novel, methodologically rigorous findings that within-person elevations in current anxiety and shame – but not average levels – improve prediction of suicide desire (concurrent and prospective) and intent (concurrent) in BDD, beyond baseline clinician risk assessment alone.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Declaration of Interest

HW: Dr. Weingarden received salary and research support from the National Institute of Mental Health (NIMH; K23MH119372) for this study. Dr. Weingarden receives salary support from HabitAware, Inc., is a paid scientific advisor to Augmend Health, and has a paid consulting agreement with APA Labs. Dr. Weingarden has received research support from Koa Health, Inc.

AJ: none declared.

MA is a compensated member of Ilumivu's scientific advisory board and could potentially benefit from the results of this research. Dr. Arney holds stock options in a company that provides ecological momentary assessment services.

CA: none declared.

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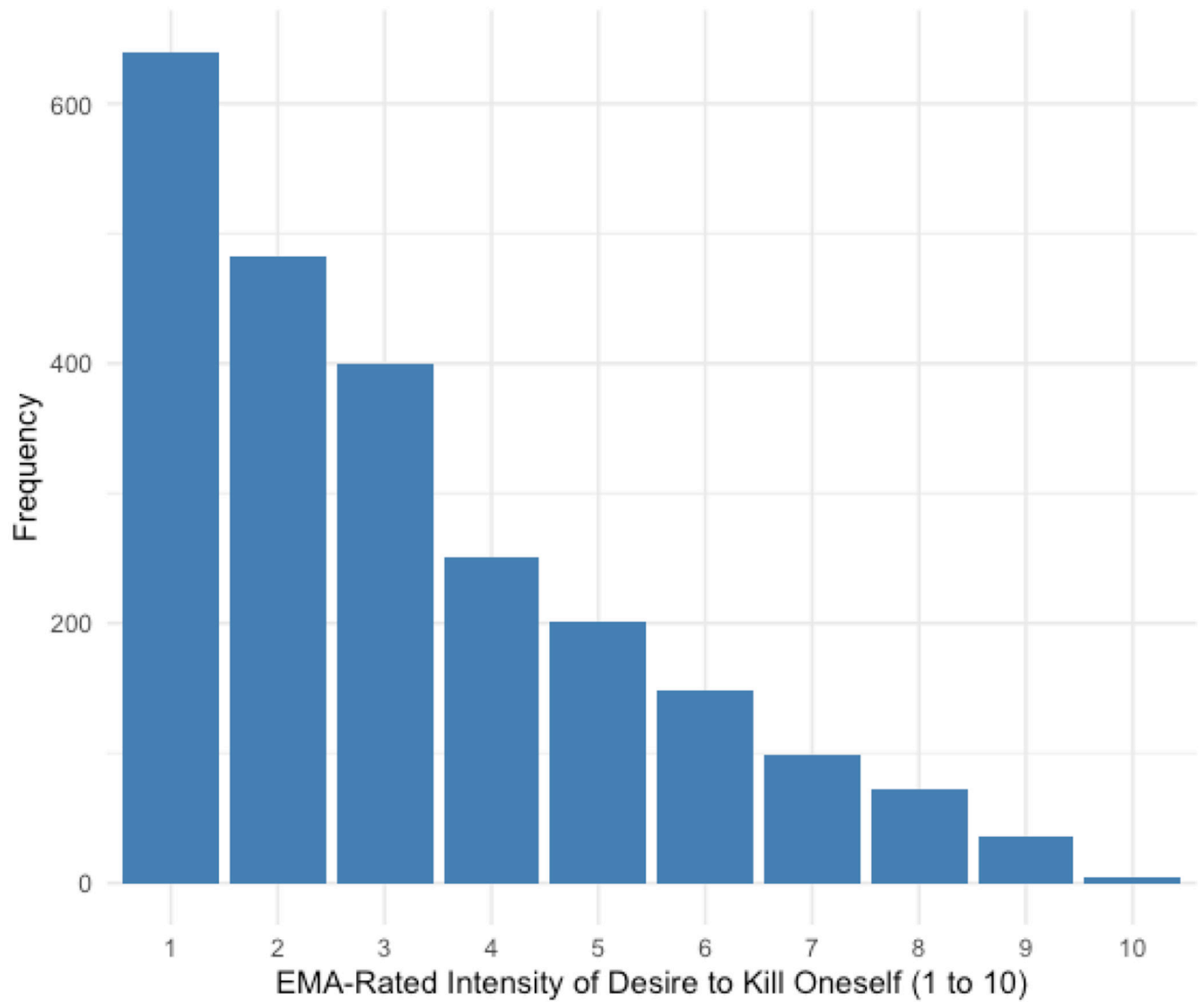


Figure 1.

Distribution of Non-Zero EMA Responses to Current Desire to Kill Oneself.

Note. Zero scores, not shown in figure, comprised 2992 (56%) of responses to this EMA item.

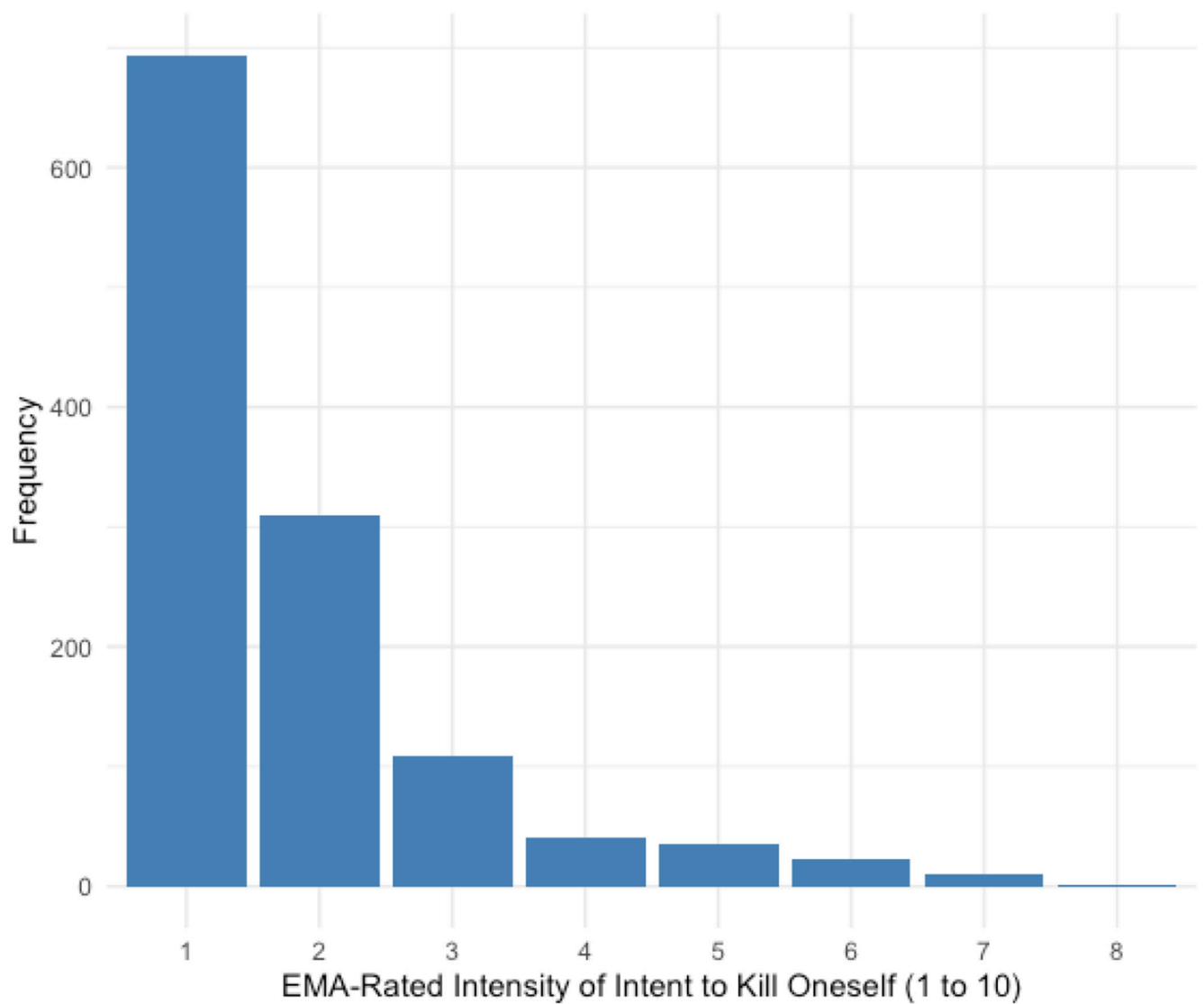


Figure 2.
Distribution of Non-Zero EMA Responses to Current Intent to Kill Oneself.
Note. Zero scores, not shown in figure, comprised 4101 (77%) of responses to this EMA item.

Table 1.
Baseline demographic and clinical characteristics of sample.

	<i>n</i> (%)
Race	
White	63 (72.41)
Black or African American	7 (8.05)
Asian	6 (6.90)
More than one race	9 (10.34)
Another race, not specified	2 (2.30)
Ethnicity	
Hispanic or Latino	13 (14.94)
Gender	
Female	64 (73.56)
Male	18 (20.69)
Genderqueer or nonbinary	3 (3.44)
Unsure or another identity	2 (2.30)
Sex at Birth	
Female	68 (78)
Male	19 (22)
Marital Status	
Single or never married	60 (68.97)
Married	13 (14.94)
Living with partner	10 (11.49)
Divorced or separated	4 (4.60)
Employment Status ^a	
Working full-time	31 (35.63)
Working part-time	25 (28.74)
Student	18 (20.69)
Unemployed	20 (22.99)
Homemaker or stay-at-home parent	6 (6.90)
Education	
High school diploma/GED or less	11 (12.64)
Some college or 2-year degree or certificate	31 (35.63)
College	28 (32.18)
Some post-graduate / professional	3 (3.45)
Post-graduate or professional degree	14 (16.09)
Living situation	
Spouse / partner / children	26 (29.89)
Parents	18 (20.69)
Roommate	21 (24.14)
Alone	16 (18.39)
Other	6 (6.90)

	<i>n</i> (%)
Current psychiatric comorbidities^{a, b}	
At least 1 comorbid diagnosis	87 (100%)
Alcohol use disorder	33 (37.93)
Substance use disorder	33 (37.93)
Major depressive disorder	32 (36.78)
Social anxiety disorder	21 (24.14)
Obsessive compulsive disorder	18 (20.69)
Generalized anxiety disorder	14 (16.09)
Agoraphobia	13 (14.94)
Panic disorder	11 (12.64)
Posttraumatic stress disorder	10 (11.49)
Taking psychotropic medication(s) at baseline	

^aSome subjects reported more than one employment status or comorbid diagnosis.

^bOnly lists disorders which have > 10% prevalence in the sample.

Table 2.
Models Predicting Concurrent Intensity of SI Desire and Presence of SI Intent

<i>Predictors</i>	Concurrent Model of SI Desire				Concurrent Model of SI Intent (Y/N)			
	<i>beta</i>	<i>Exp(b)</i>	<i>Std. CI</i>	<i>Adj. p</i>	<i>beta</i>	<i>OR</i>	<i>Std. CI</i>	<i>Adj. p</i>
C-SSRS (baseline)	0.18	1.19	1.07 – 1.33	.010	1.55	4.72	1.60 – 13.87	.045
EMA anxiety (within)	0.11	1.12	1.11 – 1.13	.010	0.45	1.58	1.38 – 1.80	.010
EMA anxiety (between)	0.13	1.13	0.99 – 1.31	.406	1.10	3.02	0.71 – 12.85	.645
EMA shame (within)	0.13	1.14	1.13 – 1.15	.010	0.48	1.61	1.43 – 1.82	.010
EMA shame (between)	0.18	1.20	1.04 – 1.39	.095	0.63	1.88	0.40 – 8.88	1.000
Marginal R ²	0.46				0.27			
Conditional R ²	0.61				0.87			
Random Effects								
σ^2	0.17				3.29			
$\tau_{00\ patid}$	0.07				16.76			
$\tau_{11\ patid.emaday}$	0.00				0.02			
$\rho_{01\ patid}$	–0.52				–0.06			
<i>ICC</i>	0.28				0.84			
<i>N</i>	87				87			

Note: SI = suicidal ideation; Std. CI = standardized confidence interval; Adj. p = p value corrected for false discovery rate; OR = odds ratio; C-SSRS = Columbia Suicide Severity Rating Scale; EMA = ecological momentary assessment; patid = participant identification number; emaday = day of EMA; ICC = intra-class correlation.

Table 3.
Models Predicting Next-Observation Intensity of SI Desire and Presence of SI Intent

<i>Predictors</i>	Prospective Model of SI Desire				Prospective Model of SI Intent (Y/N)			
	<i>beta</i>	<i>Exp(b)</i>	<i>Std. CI</i>	<i>Adj. p</i>	<i>beta</i>	<i>OR</i>	<i>Std. CI</i>	<i>Adj. p</i>
C-SSRS (baseline)	0.19	1.21	1.08 – 1.35	.010	1.51	4.54	1.55 – 13.25	0.049
EMA anxiety (within)	0.04	1.04	1.02 – 1.05	.010	0.17	1.19	1.03 – 1.37	.110
EMA anxiety (between)	0.15	1.16	1.00 – 1.34	0.260	1.05	2.87	0.67 – 12.30	0.704
EMA shame (within)	0.04	1.04	1.03 – 1.06	.010	0.06	1.06	0.93 – 1.20	1.000
EMA shame (between)	0.18	1.20	1.03 – 1.39	.110	0.59	1.80	0.39 – 8.37	1.000
Hours between observations	.08	1.08	0.91 – 1.29	1.000	2.28	9.75	1.73 – 55.10	.0738
Marginal R ²	0.37					0.25		
Conditional R ²	0.56					0.87		
Random Effects								
σ^2	0.22					3.29		
$\tau_{00\ patid}$	0.10					16.18		
$\tau_{11\ patid.emaday}$	0.00					0.02		
$\rho_{01\ patid}$	–0.58					–0.12		
<i>ICC</i>	0.31					0.83		
<i>N</i>	86					86		

Note: SI = suicidal ideation; Std. CI = standardized confidence interval; Adj. p = p value corrected for false discovery rate; OR = odds ratio; C-SSRS = Columbia Suicide Severity Rating Scale; EMA = ecological momentary assessment; patid = participant identification number; emaday = day of EMA; ICC = intra-class correlation.